Main

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Link to the github project

https://github.com/morganelpgit/data_management_project.git

Research question

How differently does quantitative easing by the ECB affect corporate borrowing costs in the eurozone, and is this difference correlated with factors such as domestic economic development and inflation?

Since its first introduction under the Obama administration as a response to the subprime mortgage crisis of 2008-2009, quantitative easing has always been labeled as an "unconventional" policy, its actual effectiveness and associated risks debated till this day. Despite its risks, the challenges faced by the eurozone pushed the European Central Bank to adopt its own version of this monetary policy: The Asset Purchase Programme (APP). Unlike the US Federal Reserve, the ECB launched its program in 2015, years after the acute phase of the sovereign debt crisis, primarily to combat deflationary risks and, critically, to counteract the severe financial fragmentation that had developed between core and peripheral member states (an important example would be Greece and Germany).

There is ample research surrounding quantitative easing, much of which focuses on Europe, examining the impact on sovereign bond yields, stock market behavior, and the overall macroeconomic aggregates like inflation and GDP growth. However, we observe fairly little research on the private sector that extends beyond the financial market – specifically, there isn't a great amount of research on the direct effect of quantitative easing on corporations' day-to-day operations. This is an important question because one of the fundamental functions of any monetary policy is, after all, to regulate economic development to ensure a healthy, efficient economy, and the operations of private enterprises are the driving force of the economy. Therefore, we would like to investigate the effect of quantitative easing on one key aspect of the private sector: the funding of corporations. In addition, from the existing literature, we were able to discover interesting studies focusing on country-level effects of the APP. Such research allowed us to realize that, even though the economic chasm between certain countries of Europe – highlighted by the Greek debt crisis of 2010 – was what pushed the ECB to adopt quantitative easing, that same chasm would also produce unexpected effects of quantitative easing.

Therefore, we propose the following research question:

How does quantitative easing by the ECB affect corporate borrowing costs in the eurozone (the European Monetary Union), and how different is this impact across different members of the EMU?

We will also attempt to observe whether this difference correlated with factors such as domestic economic development and inflation.

This is a two-part inquiry that investigates both the transmission of unconventional monetary policy and the asymmetry of this transmission. The goal is to move beyond the study of the eurozone as a monolith, to examine its fragmentation, and how that fragmentation complicates decisions made by a central, unifying authority like the ECB.

To limit our research only to countries directly regulated by the European Central Bank, we have selected data only from the EMU. Sourcing data from publicly accessible websites such as the OECD and the ECB Data Portal, we will first run a regression analysis of corporate borrowing costs in the EMU against monthly asset purchases by the ECB (incorporating lags and proxies/instruments to represent generale economic policy tightness/looseness) to establish a benchmark coefficient for the EMU. Then, we will estimate country-specific coefficients and

conduct cluster analysis (e.g. using KNN analysis) to determine which countries display the most similar coefficients, and highlight the economic features that define countries in the same cluster.

Link to the sources

- Quantitative Easing data : We are using the csv file "APP_breakdown_history" (click on the link "History of cumulative purchase breakdowns under the APP") downloaded from the webpage dedicated to the Asset Purchase Programme : https://www.ecb.europa.eu/mopo/implement/app/html/index.en.html).

For all other datasets of the ECB Data Portal, we browsed the series name through the the search engine, select the option "All MU countries (20)" in the "Geographical areas", "select all", and download as csv files that contains all countries times series.

- Cost of borrowing for corporations , Euro area, Monthly https://data.ecb.europa.eu/search-results?searchTerm=Cost%20of%20borrowing%20for%20corporations
- Direct investment, net, transactions, Monthly https://data.ecb.europa.eu/search-results?searchTerm=Direct%20investment%2C%20net%2C%20transactions%2C%20Monthly
- HICP - Overall index, Euro area, Monthly https://data.ecb.europa.eu/search-results?searchTerm=HICP%20-%20Overall%20index%2C%20Euro%20area%2C%20Monthly
- Euribor 3-month - Historical close, average of observations through period, Euro area, Monthly https://data.ecb.europa.eu/data/datasets/FM/FM.M.U2.EUR.RT.MM.EURIBOR3MD/_.HSTA
- Euribor 6-month - Historical close, average of observations through period, Euro area, Monthly https://data.ecb.europa.eu/data/datasets/FM/FM.M.U2.EUR.RT.MM.EURIBOR6MD/_.HSTA
- Long-term interest rate for convergence purposes – 10-year maturity – denominated in euro – Euro area 20 (fixed composition) https://data.ecb.europa.eu/search-results?searchTerm=Long-term%20interest%20rate%20for%20convergence%20purposes%20%E2%80%93%2010-year%20maturity%20%E2%80%93%20denominated%20in%20euro

The ECB Data Portal https://data.ecb.europa.eu/ is the European Central Bank's online delivery service for statistical data and related metadata and publications.

Description of the sources

Quantitative Easing variables : the Asset Purchase Programme (2014)

- Quantitative Easing (QE) Shocks (Source: APP_breakdown_history.csv from the ECB website): The monthly net amount in euro millions of ECB asset purchases. This dataset provides the breakdown of cumulative net purchases under the Asset Purchase Programme (APP) - the ECB's quantitative easing programme - covering the period 2014–2025. We focus on the first four columns of the dataset, which report monthly net purchases at book value for each of the four components of the APP, which are the corporate sector purchase programme (CSPP), the public sector purchase programme (PSPP), the asset-backed securities purchase programme (ABSPP) and the third covered bond purchase programme (CBPP3).

Target variable : Corporate borrowing rates

- Corporate borrowing rates (Source: ECB data portal) : This is the weighted average annual interest rates of new loans for corporations in each country.

Other variables

- Direct investment, net, transactions, Euro Area 20, Monthly (Source: ECB Data Portal): Net inflows and outflows of direct investment for each country. This variable helps to assess whether quantitative easing have caused capital flight between core and peripheral Euro area countries.
- Harmonised Index of Consumer Prices (HICP) (Source: ECB data portal). We will use this variable to compute the real inflation rate for each country.
- Short term rates : Euribor 3-month & 6-month - Historical close, average of observations through period, Euro area, Monthly (Source: ECB data portal). Euribor is determined by the ECB's monetary policy. These values are the average interest rates in the interbank market, and serve as a benchmark for interest rates to all non-financial agents (therefore more general than the corporate borrowing rate, our target variable). These nominal rates are common across all Euro area. We will then compute the real interest rate for each country. Using the Fisher approximation, the real interest rate is obtained by subtracting a country's inflation rate from the nominal interest rate.
- Long-term interest rate – 10-year maturity – denominated in euro – Euro area 20 (Source: ECB data portal). This variable represents the yield on government bonds, and helps us to assess whether QE has had different effects on core and peripheral Euro area countries.

Set-up

```
here::i_am("data_management_project.Rproj")

library(tidyverse)
library(here)
library(vroom)
library(dplyr)
library(tidyr)
library(stringr)
library(ggplot2)
```

Data Loading & Data Cleaning

Quantitative Easing variable : the Asset Purchase Programme (2014)

Load Data: Historical data of asset puchases by the ECB & Data cleaning

Data cleaning: Remove useless data & Adding dates & Converting variables to proper formats

```
#We load the .csv file directly downloaded from the ECB website titled "APP_breakdown_history
QE_mnet <- vroom(here("data", "APP_breakdown_history.csv"))

QE_init_col <- ncol(QE_mnet)
QE_init_row <- nrow(QE_mnet)
QE_init_na <- sum(is.na(QE_mnet))

#The title row and first two rows of the original .csv file are empty except for some textual

#First, we remove the first two rows
QE_mnet <- QE_mnet |>
  slice(-(1:2))|>
  slice(-(135:137))
#Second, the names of the variables need to be manually added.

names(QE_mnet)[1] <- paste("Year")
names(QE_mnet)[2] <- paste("Month")
names(QE_mnet)[3] <- paste("ABSPP") #Asset-backed securities purchase programme
names(QE_mnet)[4] <- paste("CBPP3") #Covered bond purchase programme 3
names(QE_mnet)[5] <- paste("CSPP") #Corporate Sector purchase programme
names(QE_mnet)[6] <- paste("PSPP") #Public Sector purchase programme
```

```r
#All data starting from the 7th column describe quarterly adjustments and total holdings, wh
QE_mnet <- QE_mnet|>
  dplyr::select(1:6)

#In the original .csv, the year column is empty for all months except for the first month of


#First, we add the missing years.

QE_mnet <- QE_mnet |>
  fill(Year, .direction = "down")

#Second, we manually create a date column titled "DATE" (because later we will merge with ot

QE_mnet <- QE_mnet |>
  mutate(
    Month_num = match(Month, month.name),
    Month_num = ifelse(is.na(Month_num), match(Month, month.abb), Month_num),
    date_first = make_date(year = Year, month = Month_num, day = 1),
    last_day = days_in_month(date_first),
    DATE = paste0(Year,"/", Month_num, "/", last_day)
  ) |>
  dplyr::select(-Month_num, -date_first, -last_day)


QE_mnet <- QE_mnet|>
  mutate(DATE = as.Date(DATE, format = "%Y/%m/%d"))


#Last, since we are only interested in the magnitude of quantitative easing carried out by t

#First, we convert them from character to numerical format
QE_mnet <- QE_mnet|>
  mutate(ABSPP = as.numeric(ABSPP),
         CBPP3 = as.numeric(CBPP3),
         CSPP = as.numeric(CSPP),
         PSPP = as.numeric(PSPP))
#Second, we add them together
QE_mnet <- QE_mnet|>
  mutate(Mnet = ABSPP+CBPP3+CSPP+PSPP)
```

	Before cleaning	After cleaning
Number of columns	14	8
Number of rows	139	134
Number of NAs	174	0

Target variable : corporate borrowing rates

Loading Data: Corporate borrowing rates per country (downloaded from the ECB data portal)

Data cleaning: Renaming variables & Pivoting

```
corp_int <- vroom(here("data","corp_int.csv"))

corp_init_col <- ncol(corp_int)
corp_init_row <- nrow(corp_int)
corp_init_na <- sum(is.na(corp_int))

#The original .csv file is in wide format, with one column for each country. The problem is t

corp_int <- corp_int |>
  rename_with(~ ifelse(
    str_detect(., "MIR\\.M\\.[A-Z]{2}\\."),
    str_extract(., "(?<=MIR\\.M\\.)[A-Z]{2}(?=\\.)"),
    .
  ))


#The previous function does not the code "U2" (which represents the EU) because it does not c
names(corp_int)[23] <- paste("U2")

#We change the date format, remove the useless "TIME PERIOD" column and filter the useful per
corp_int <- corp_int|>
  mutate(DATE = as.Date(DATE, format = "%Y/%m/%d"))

corp_int <- corp_int|>
  dplyr::select(!`TIME PERIOD`) |>
  filter(DATE >= as.Date("2014-10-01") & DATE <= as.Date("2025-11-30"))

#We pivot from long to wide, so that now we have observations in country-months
corp_int <- corp_int|>
```

```
    pivot_longer(
    cols = -DATE,
    names_to = "Country",
    values_to = "int"
  )
```

	Before cleaning	After cleaning
Number of columns	23	3
Number of rows	277	2772
Number of NAs	800	102

The remaining NAs originate from missing data for Hungary (HR) and three missing observations for Lithuania (LT). We will eventually remove this country's values from the analysis.

First merge

Merging corporate interest rates & QE datasets

```
qe_int <- left_join(QE_mnet,corp_int, by = join_by(DATE)
  )
data1 <- qe_int
```

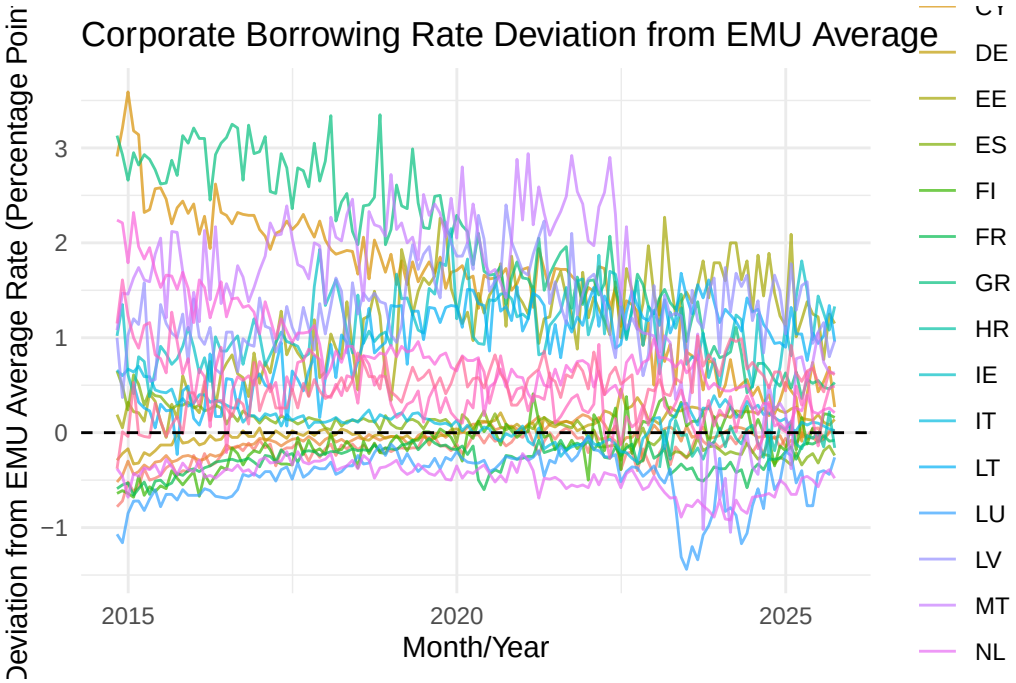
Graphical representation of target variable

```
#Preliminary visusalization of varying differences between corporate borrowing rates of each

emu_avg_rates <- corp_int |>
  drop_na() |>
  filter(Country == "U2") |>
  dplyr::select(DATE, EMU_Avg = int)

corp_int_deviation <- corp_int |>
  drop_na() |>
  filter(Country != "U2") |>
  left_join(emu_avg_rates, by = "DATE") |>
  mutate(Deviation = int - EMU_Avg) |>
  dplyr::select(-EMU_Avg)
```

```r
corp_int_deviation|>
  ggplot(aes(x = DATE, y = Deviation, group = Country, color = Country)) +
  geom_line(alpha = 0.7) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "black") +
  labs(
    title = "Corporate Borrowing Rate Deviation from EMU Average",
    x = "Month/Year",
    y = "Deviation from EMU Average Rate (Percentage Points)"
  ) +
  theme_minimal()
```




```r
#Preliminary visualization of the cross-country differences between their relationships with

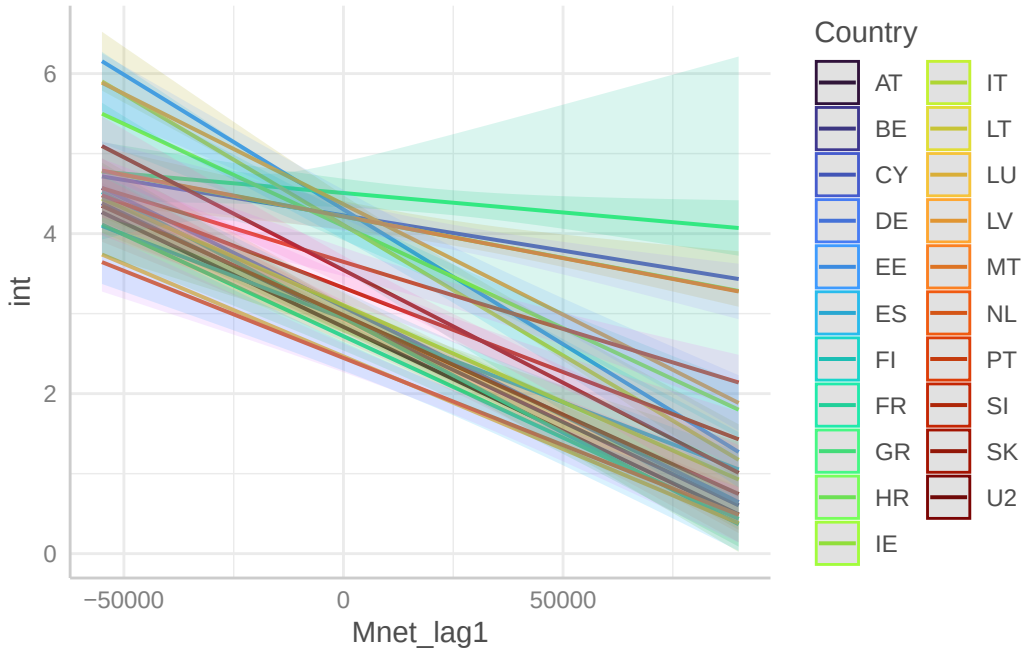
library(ggeffects)

qe_int <- qe_int |>
  arrange(Country, DATE)

qe_int <- qe_int |>
  group_by(Country) |>
  mutate(Mnet_lag1 = lag(Mnet, 1)) |>
  ungroup()
```

```r
qe_nona <- drop_na(qe_int)

fit = lm(formula = int ~Mnet_lag1*Country, data=qe_nona)
pred = ggpredict(fit, terms = c("Mnet_lag1","Country"))
plot(pred) +
  scale_color_viridis_d(option = "turbo") +
  ggtitle(NULL)
```



A few comments on data loading and cleaning

Each data file downloaded from the ECB Data Portal is loaded and cleaned as follows:

We start by loading the dataset from our data folder, and computing the original number of rows and columns. We use the function spec() with the col_types() argument to preserve the date format for `DATE`, and the numeric type for variables that are incorrectly recognized as logical.

Two columns are for the date : `DATE` as a date,`TIME PERIOD` as a character. The others are the values for the twenty countries of the Euro Union. Each of these column corresponds to a full ECB series name, for example: `Financial account, Direct Investment, Net (BPS.M.N.IE.W1.S1.S1.T.N.FA.D.F5._Z.EUR._T._X.N.ALL)`. We rename each column by extracting the two letter country code from the series name, using the function ifelse.

Next, we select all columns except `TIME PERIOD` (we will use `DATE` for joining). We also filter the observations from October 2014 to November 2025.

Finally, we transform the dataset from wide format (months as rows, countries as columns) to long format (months and countries as rows), and left join to the full `data` dataset.

Net investments

```
# We load the dataset and compute the initial number of rows and columns
invest <- vroom(here("data","Net_direct_investments.csv"),
                col_types = cols(
  DATE = col_date(format = ""),
  `TIME PERIOD` = col_character(),
  `Financial account, Direct Investment, Net (BPS.M.N.IE.W1.S1.S1.T.N.FA.D.F5._Z.EUR._T._X.N
  `Financial account, Direct Investment, Net (BPS.M.N.NL.W1.S1.S1.T.N.FA.D.F5._Z.EUR._T._X.N
))

invest_init_col <- ncol(invest)
invest_init_row <- nrow(invest)
invest_init_na <- sum(is.na(invest))


# We rename the columns of dataset
invest <- invest |>
  rename_with(~ ifelse(
    str_detect(., "BPS\\.M\\.N\\.[A-Z]{2}\\.W1"),
    str_extract(., "(?<=BPS\\.M\\.N\\.)[A-Z]{2}(?=\\.)"),
    .
  ))

# We select the variables and filter the date
invest <- invest |>
  dplyr::select(!`TIME PERIOD`) |>
  filter(DATE >= as.Date("2014-10-01") & DATE <= as.Date("2025-11-30"))

# We pivot to long format
invest <- invest |>
    pivot_longer(
    cols = -DATE,
    names_to = "Country",
    values_to = "Invest"
  )
```

	Before cleaning	After cleaning
Number of columns	22	3
Number of rows	417	2640
Number of NAs	4399	495

We decided to keep a complete dataset and select all the countries, resulting in a lot of missing values even after cleaning (AT, CY, IE, NL have no time series available). We will eventually delete these countries values for the analysis.

We then join this dataset to the previous one.

```
data2 <- left_join(data1,invest,by=join_by(DATE,Country))
```

Harmonised index of consumer prices (HICP)

```
# We load the dataset and compute the initial number of rows and columns
HICP <- vroom(here("data","HICP.csv"),
              delim = ",") #correct the problem of delimitation

HICP_init_col <- ncol(HICP)
HICP_init_row <- nrow(HICP)
HICP_init_na <- sum(is.na(HICP))

# We rename the columns
HICP <- HICP |>
  rename_with(~ ifelse(
    str_detect(., "ICP\\.M\\.[A-Z]{2}\\."),
    str_extract(., "(?<=ICP\\.M\\.)[A-Z]{2}(?=\\.)"),
    .
  ))

# We select and pivot to long format
HICP <- HICP |>
  dplyr::select(!`TIME PERIOD`) |>
  filter(DATE >= as.Date("2014-10-01") & DATE <= as.Date("2025-11-30"))

HICP <- HICP |>
```

```
  pivot_longer(
    cols = -DATE,
    names_to = "Country",
    values_to = "HICP"
  )

# We join to the previous dataset
data3 <- left_join(data2, HICP, by=join_by(DATE,Country))
```

6 months maturity

	Before cleaning	After cleaning
Number of columns	22	3
Number of rows	359	2680
Number of NAs	83	0

Short term interest rates

These two nominal rates (maturity 3 and 6 month) are common across all euro area countries and are set by the ECB. We will then compute the real interest rate for each country.

```
# We load the short term interest rate of maturity 3 months, and compute the initial number o
ST3M_int <- vroom(here("data","Euribor_3M.csv"))

ST3M_int_init_col <- ncol(ST3M_int)
ST3M_int_init_row <- nrow(ST3M_int)
ST3M_int_init_na <- sum(is.na(ST3M_int))

# This dataset only has 3 rows
colnames(ST3M_int) <- c("DATE","TIME PERIOD","ST3M")

ST3M_int <- ST3M_int |>
  dplyr::select(DATE,ST3M) |>
  filter(DATE >= as.Date("2014-10-01") & DATE <= as.Date("2025-11-30"))

# We load the short term interest rate of maturity 6 months, and compute the initial number o
ST6M_int <- vroom(here("data","Euribor_3M.csv"))

ST6M_int_init_col <- ncol(ST6M_int)
ST6M_int_init_row <- nrow(ST6M_int)
```

```
ST6M_int_init_na <- sum(is.na(ST6M_int))

# This dataset only has 3 rows
colnames(ST6M_int) <- c("DATE","TIME PERIOD","ST6M")

ST6M_int <- ST6M_int |>
  dplyr::select(DATE,ST6M) |>
  filter(DATE >= as.Date("2014-10-01") & DATE <= as.Date("2025-11-30"))

data4 <- left_join(data3,ST3M_int,by="DATE")

data4 <- left_join(data4,ST6M_int,by="DATE")
```

3 months maturity

	Before cleaning	After cleaning
Number of columns	3	2
Number of rows	383	134
Number of NAs	0	0

There are less rows after cleaning because of all the dates removed.

6 months maturity

	Before cleaning	After cleaning
Number of columns	3	2
Number of rows	383	134
Number of NAs	0	0

Long term interest rate

We import the long term interest rate for 10 years maturity. As well as the columns "Date", "Time Period" and the country values, there are two aggregates of code I9 (Euro area 20 (fixed composition) as of 1 January 2023) and U2 (Euro area changing composition), hence 24 columns.

```
# Load and compute
LT10Y_int <- vroom(here("data","LT_interest_rate.csv"))
```

```r
LT10Y_int_init_col <- ncol(LT10Y_int)
LT10Y_int_init_row <- nrow(LT10Y_int)
LT10Y_int_init_na <- sum(is.na(LT10Y_int))

# We rename the columns
LT10Y_int <- LT10Y_int |>
  rename_with(~ ifelse(
    str_detect(., "IRS\\.M\\.[A-Z0-9]{2}\\."),
    str_extract(., "(?<=IRS\\.M\\.)[A-Z0-9]{2}(?=\\.)"),
    .
  ))

# We select and pivot to long format
LT10Y_int <- LT10Y_int |>
  dplyr::select(DATE, AT, BE, CY, DE, EE, ES, FI, FR, GR, HR, IE, IT, LT, LU, LV, MT, NL, PT
  ) |> # we don't select I9
  filter(DATE >= as.Date("2014-10-01") & DATE <= as.Date("2025-11-30"))

LT10Y_int <- LT10Y_int |>
  pivot_longer(
    cols = -DATE,
    names_to = "Country",
    values_to = "LT10Y"
  )

# We join to the final dataset
data <- left_join(data4, LT10Y_int, by = join_by(DATE, Country))
```

	Before cleaning	After cleaning
Number of columns	24	3
Number of rows	478	2793
Number of NAs	2709	69

We still observe many NAs because Estonia is included, and its data only start in 2020.